

Prime Palindrome

Sample Case Study on Classes & Objects – CO1

1. Aim

To design and implement a Java program using a class and object to check whether a given positive integer is both a prime number and a palindrome.

2. Problem Statement

Write a Java program that accepts a number from the user and checks whether it is a prime palindrome. The program first checks whether the number is prime. If it is prime, it reverses the number and compares the reversed value with the original value to determine whether it is a palindrome.

3. Concepts / Class and Object Used

Component	Description
PrimePalindrome	Main class containing the program logic.
checkprime(int num)	Checks whether the number is prime.
check(int n)	Reverses the number and checks whether it is a palindrome.
Scanner	Reads the number entered by the user.
Object p	Object of PrimePalindrome used to call checkprime().

4. Algorithm

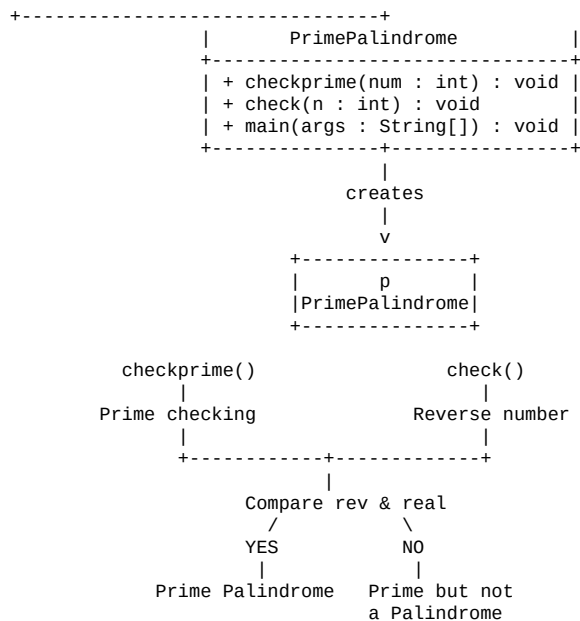
1. Start the program.
2. Create a Scanner object to read an integer.
3. Create an object p of the PrimePalindrome class.
4. Call checkprime(number).
5. If the number is zero or negative, display an invalid-input message.
6. If the number is 1, display that it is not a prime palindrome.
7. For a number greater than 1, check divisibility from 2 up to num - 1.
8. If a divisor is found, the number is not prime.
9. If no divisor is found, call check(num).
10. Reverse the number using % and / operations.
11. Compare the reversed number with the original number.
12. If equal, display that the number is a prime palindrome; otherwise display that it is prime but not a palindrome.
13. Stop the program.

5. Execution Steps

1. Open Eclipse IDE and create a Java project.
2. Create the package named observation.
3. Create a Java class named PrimePalindrome.

4. Enter the given program.
5. Compile the program.
6. Run the PrimePalindrome class.
7. Enter an integer when prompted.
8. Observe the result in the console.
9. For the sample output, enter 131.

6. UML Class Diagram



7. Dry Run / Process

Step	Operation	Result
1	User enters number	131
2	Check divisibility by 2 to 130	No divisor found
3	Prime check	131 is prime
4	Reverse 131	131
5	Compare reverse and original	131 == 131
6	Final result	Prime palindrome

8. Complete Java Program

```

package observation;
import java.util.Scanner;

public class PrimePalindrome {
    void checkprime(int num) {
        boolean x=true;
        if(num<0 || num==0) {
            System.out.println("the check for ZER or negative INVALID");
        }
        else if(num==1) {
            System.out.println("The number is not a Prime Palindrome");
            return;
        }
        else {
            for(int i=2;i<num;i++) {
                if(num%i==0) {
                    System.out.println("the number is not a primepalindrome");
                    x=false;
                    break;
                }
            }
        }
        if(x) {
            check(num);
        }
    }

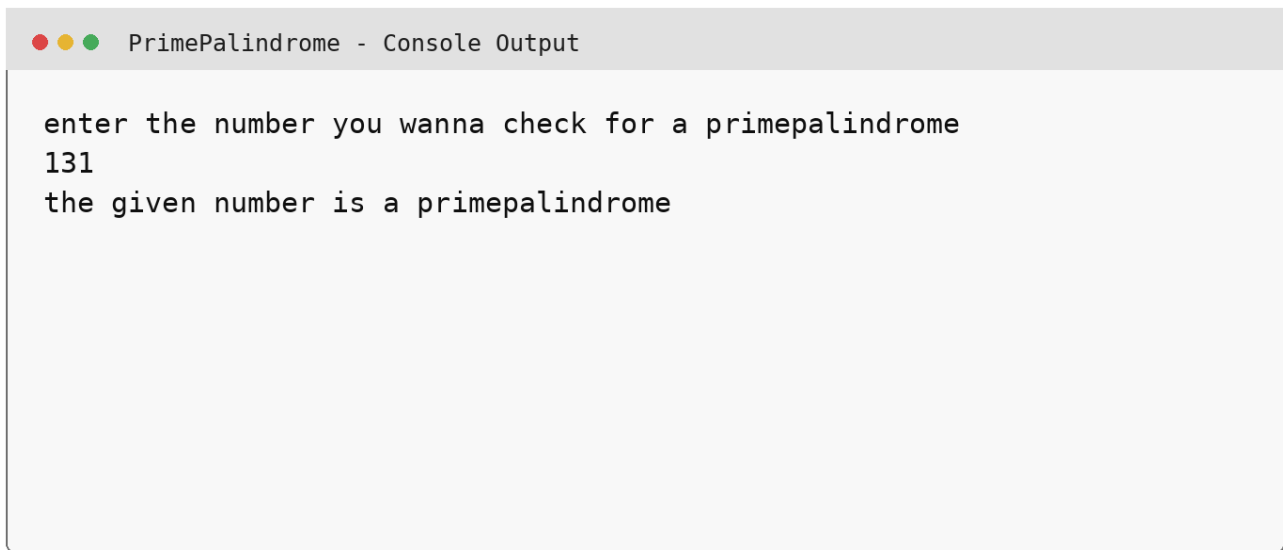
    void check(int n) {
        int real=n;
        int rev=0,r;
        while(n>0) {
            r=n%10;
            n=n/10;
            rev=rev*10+r;
        }
        if(rev==real) {
    
```

```
        System.out.println("the given number is a primepalindrome");
    }
    else {
        System.out.println("tehe number is prime but not a palindrome");
    }
}

public static void main(String[] args) {
    int number;
    Scanner sc=new Scanner(System.in);
    System.out.println("enter the number you wanna check for a primepalindrome");
    number =sc.nextInt();
    PrimePalindrome p=new PrimePalindrome();
    p.checkprime(number);
}
}
```

9. Program Output – Screenshot

Sample execution with input 131:



```
enter the number you wanna check for a primepalindrome
131
the given number is a primepalindrome
```

10. Result / Conclusion

The Prime Palindrome program was successfully implemented using a Java class and object. It first checks whether the given number is prime and then checks whether the prime number is a palindrome. For input 131, the program successfully identifies it as a prime palindrome.